

AI FREQUENTLY ASKED QUESTIONS (FAQ) ASSISTANT API

PROJECT DOCUMENTATION

Team ID	SWTID-2026-1039
Team Size	5
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Naan Mudhalvan / TN Skills Project

Documentation covering the purpose, workflow, API demonstration, features, testing, benefits, limitations, and future scope of the project.

Prepared by the Project Team

1. Project Overview

The AI Frequently Asked Questions (FAQ) Assistant API is a backend-based application designed to receive user questions and return AI-generated answers through an API. It demonstrates a simple question-and-answer workflow in which a structured request is sent to the backend and a structured response is returned.

2. Problem Statement

Users often need quick answers to frequently asked questions. Searching manually can take time, particularly when similar questions are asked repeatedly. This project addresses the need by providing an API-based assistant that can receive a question and generate a relevant response.

3. Objectives

- Provide an API interface for submitting frequently asked questions.
- Process user questions through a backend application.
- Generate meaningful AI-assisted answers.
- Return the result in a structured response format.
- Demonstrate successful request handling using HTTP status information.

4. Scope of the Project

The project focuses on the API layer and the interaction between a user question and the generated answer. It can serve as a foundation for FAQ systems in educational portals, service applications, help desks, websites, and other information-oriented systems.

5. System Workflow

The basic workflow of the AI FAQ Assistant API is:

Step	Process
1	User enters a question.
2	The question is placed in a JSON request body.
3	A POST request is sent to the AI answer endpoint.
4	The backend receives and processes the request.
5	The AI component generates a suitable answer.
6	The API sends the result back to the requester.

6. Main Components

The demonstrated backend follows a modular logical structure:

- **Routes:** Define the API endpoints through which requests are received.
- **Controllers:** Handle incoming requests and coordinate the response process.
- **Services:** Contain processing logic required to generate or obtain the answer.
- **Models:** Provide a structured place for application data when required.
- **Middleware:** Supports request processing and common backend operations.

7. API Demonstration

The demo uses a POST request to the AI answer endpoint. The request body contains a question in JSON format. The example question used in the demonstration is **“What is MongoDB?”**

After sending the request, the API returns a **200 OK** status, indicating that the request was successfully processed. The response contains the generated answer.

8. Example API Interaction

The following summarizes the demonstrated request:

Request Method	POST
Endpoint	/api/ai/answer
Input Format	JSON
Example Question	What is MongoDB?

Response

The API responds successfully with a 200 OK status and returns an AI-generated explanation. For the MongoDB example, the response explains MongoDB as a popular NoSQL database system that stores data in flexible, JSON-like documents rather than traditional table-and-row structures.

9. Key Features

- API-based question submission.
- Structured JSON request and response handling.
- AI-assisted answer generation.
- Clear success status through HTTP response codes.
- Modular backend design that can be extended.
- Potential integration with websites, mobile applications, or other client interfaces.

10. Benefits

- Reduces the effort required to obtain answers to common questions.
- Provides a consistent API interface for client applications.
- Can support faster information retrieval.
- Can be expanded to support domain-specific FAQs.
- Demonstrates practical integration of AI capabilities into an application backend.

11. Testing and Validation

The API can be tested by sending POST requests with different questions and checking both the HTTP status and returned response. A successful 200 OK response confirms that the request was accepted and processed by the API.

Test Case	Expected Result
Valid FAQ question	API processes the request and returns an answer.
Valid JSON request	Request is accepted by the endpoint.
Successful processing	200 OK response is returned.
Different questions	Assistant generates responses based on the submitted question.

12. Limitations

The quality of an AI-generated response depends on the input question and the underlying AI service. The system should therefore be tested with different types of questions, and responses should be reviewed when the application is used for important or sensitive information.

13. Future Enhancements

- Add authentication and authorization for protected API access.
- Add logging and monitoring for API requests and errors.
- Connect the API to a user-friendly web or mobile interface.
- Support conversation history and contextual follow-up questions.
- Add domain-specific knowledge sources for more controlled FAQ responses.
- Improve validation and error messages for invalid requests.

14. Conclusion

The AI FAQ Assistant API demonstrates how an AI-powered question-and-answer service can be exposed through a backend API. A user submits a question in a structured request, the backend processes it, and the system returns an AI-generated answer with a successful response status. The project provides a practical foundation for intelligent FAQ and support applications and can be extended with additional security, user-interface, data, and AI capabilities.

15. Team Details

Project team for the Naan Mudhalvan / TN Skills submission:

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Project Topic

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